

Human-Assisted Cognitive Defect Inspector (HCDI)

- AI-Powered Visual Defect Detection System
 - Artificial Intelligence Capstone Project
 - Abhishek Patnaik

Problem Statement

- Manual inspection is time-consuming and inconsistent.
 - Human fatigue can reduce defect detection accuracy.
 - Need for automated intelligent inspection systems.

Project Objective

- Develop CNN-based defect classification model.
 - Estimate defect severity quantitatively.
 - Generate human-readable AI inspection remarks.

System Architecture

- Image Input → Preprocessing → CNN Feature Extraction
 - Binary Classification (Defect / No Defect)
 - Severity Estimation & AI Remark Generation

Technology Stack

- Python
 - Google Colab
 - TensorFlow / Keras
 - OpenCV
 - NumPy & Matplotlib

Unique Dataset Creation

- Synthetic dataset generated programmatically.
 - Defect samples contain artificial irregular regions.
 - Ensures originality and avoids external dataset usage.

Image Preprocessing

- Resizing images to 128×128 pixels.
 - Normalization using rescale = 1/255.
 - Training-validation split with ImageDataGenerator.

CNN Model Design

- Convolution layers extract spatial features.
 - MaxPooling layers reduce dimensions.
 - Dense layers perform final binary classification.

Model Training Strategy

- Binary Cross-Entropy loss function.
 - Adam optimizer for faster convergence.
 - Model trained over multiple epochs.

Inspection Intelligence Engine

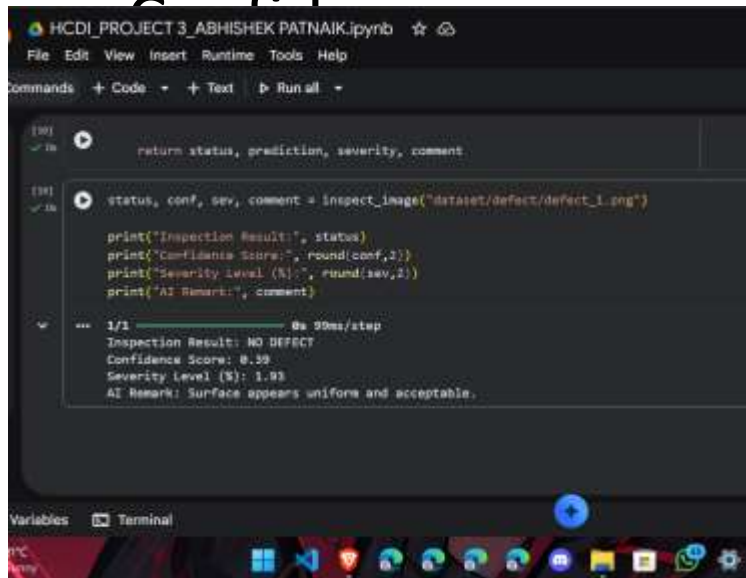
- Predicts probability of defect presence.
 - Calculates confidence score.
 - Computes severity level using pixel intensity ratio.

Severity Estimation Mechanism

- Dark pixel concentration analysis.
 - Higher irregular region → Higher severity score.
 - Provides quantitative defect percentage.

Sample Inspection Output

- Model outputs classification result, confidence score, severity percentage, and AI



The screenshot shows a Jupyter Notebook interface with a dark theme. The notebook is titled 'HCDI_PROJECT 3_ABHISHEK PATNAIK.ipynb'. The code cell contains the following Python code:

```
[38]: return status, prediction, severity, comment

[39]: status, conf, sev, comment = inspect_image("dataset/defect/defect_1.png")

print("Inspection Result:", status)
print("Confidence Score:", round(conf,2))
print("Severity Level (%):", round(sev,2))
print("AI Remark:", comment)
```

The output of the code cell is displayed below the code:

```
1/1 ----- 0s 90ms/step
Inspection Result: NO DEFECT
Confidence Score: 0.99
Severity Level (%): 1.93
AI Remark: Surface appears uniform and acceptable.
```

The bottom of the notebook shows the 'Variables' and 'Terminal' tabs, and a Windows taskbar is visible at the very bottom.

Project Implementation Link

- Google Colab Notebook:
 - https://colab.research.google.com/drive/12uFYvPwUzOld_urd5LEO7TZ_8Tjv5fhl?usp=sharing

Applications

- Manufacturing Quality Control
 - Automotive Component Inspection
 - Electronics PCB Testing
 - Industrial Automation Systems

Conclusion & Future Scope

- Automates industrial inspection tasks efficiently.
 - Reduces human error in quality evaluation.
 - Future: Integrate Grad-CAM explainability & deployment.